MILITARY SPECIFICATION SHEET

SCANNER, MECHANICAL, MX-10717(V)1/U

This specification is approved for use by USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification sheet covers the Mechanical Scanner, MX-10717(V)1/U and augments MIL-S-49301. The MX-10717(V)1/U is used on the AN/ASQ-170, Target Acquisition and Designation Sight; and the AN/AAQ-11, Pilot Night Vision Sensor (TADS/PNVS).

1.2 Purpose. The purpose of this specification sheet is to modify or expand the contents of MIL-S-49301 to describe the MX-10717(V)1/U Mechanical Scanner. All the MIL-S-49301 contents apply except as specifically modified herein. Paragraph numbers directly correlate between MIL-S-49301 and this specification sheet.

2. APPLICABLE DOCUMENTS

2.1 Order of precedence. In the event of a conflict between the text of this specification sheet and the references cited herein, the text of this specification sheet shall take precedence.

SPECIFICATIONS

MILITARY

MIL-S-49301	- Scanner, Mechanical,MX-10717(V)/U, General Specification for

DRAWINGS

USA COMMUNICATIONS-ELECTRONICS COMMAND

54490 5006243-1	- Scanner, Mechanical, MX-10717(V)1/U
80063 SM-D-773946	- Scan-Interlace, PL-1408/UA

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth NJ 07703, by using the self-addressed Standardization Document Improvement proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:02Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Description. The MX-10717(V)1/U Scanner, herein known as the MX-10717(V)1/U shall consist of the MIL-S-49301, MX-10717(V)1/U Scanner adjusted for 60 Hz operation.

3.2 Construction. The MX-10717(V)1/U shall be constructed in accordance with drawing 5006243-1.

3.2.1 Weight. Weight of the 10717(V)1/U shall be 1.45 pounds maximum.

3.6 Performance characteristics. The characteristics specified herein shall apply as constructed (3.2). Operation shall be at 60 Hz using the Scan-Interlace, PL-1408/UA (SM-D-773946).

3.6.1 Scan angle and frequency.

3.6.1.1 Variable scan frequency. Applicable for first article test only. See MIL-S-49301.

3.6.1.2 Specific scan frequency (60 Hz). Scan frequency of the MX-10717(V)1/U shall be adjusted to 60.0, ±1.0 Hz. Scan frequency drift shall not be greater than ±6.0 Hz from the frequency at ambient setting (see MIL-S-49301, 4.4) when exposed to high and low temperature environmental conditions (see MIL-S-49301).

3.6.1.3 Synchronous scan frequency. The MX-10717(V)1/U shall synchronize with a stable external synchronization source, such as the TV camera vertical sync pulse. Scan frequency shall be stable within 1 Hz of the input sync pulse frequency.

3.6.1.4 Scan angle. The scan angle shall be adjusted to 13.0, ±0.3 degrees.

3.6.2 Interlace. The MX-10717(V)1/U shall operate in a 2:1 interlace mode. The interlace switching time shown in figure 1 of MIL-S-49301 shall not be greater than 2.5 milliseconds (ms).

3.6.3 Scan direction. The MX-10717(V)1/U shall show the waveform shapes and phase relationships in figure 1 while operating in either the 2:1 side or the 2:1 rear interlace mode.

3.6.4 Power consumption. Power required to operate the MX-10717(V)1/U shall not exceed 7.5 watts.

3.6.5 Scan efficiency. The MX-10717(V)1/U scan efficiency shall be a minimum of 70 percent.

3.6.6 Scan jitter in azimuth. Jitter time shall be 12.5 microseconds maximum.

3.7 Environmental conditions. See MIL-S-49301.

Source: https://assist.dla.mil --. Downloaded: 202610-22T20:02Z
Check the source to verify that this is the currentversion before use.

4. QUALITY ASSURANCE PROVISIONS.

4.6 Test methods. Tests shall be performed utilizing the SM-D-773946 Scan-Interlace with the MX-10717(V)1/U with a scan angle of 13.0, ±0.3 degrees at the 60.0, ±1.0 Hz scan rate.

4.6.1 Scan frequency.

4.6.1.1 Variable scan frequency. See MIL-S-49301.

4.6.1.2 Specific scan frequency (60 Hz). The MX-10717(V)1/U shall be connected to the Scan-Interlace SM-D-773946, and mounted to a suitable fixture. Power shall be applied and the output of the azimuth (scan) transducer (Scan-Interlace SM-D-773946) shall be monitored with a calibrated frequency counter. The output of the Scan-Interlace (SM-D-773946) shall be adjusted to obtain the specified frequency. Failure to meet the requirements of 3.6.1.2 shall constitute failure of this test.

4.6.1.3 Synchronous scan frequency. The MX-10717(V)1/U shall be connected to Scan-Interlace SM-D-773946 and mounted in a suitable fixture. Synchronization from an external stable source (such as a TV camera vertical sync pulse) and power shall be applied and the output of the azimuth (scan) transducer (Scan-Interlace SM-D-773946) shall be monitored with a calibrated frequency counter. Failure to meet the requirements of 3.6.1.3 shall constitute failure of this test.

4.6.1.4 Scan angle. See MIL-S-49301.

4.6.2 Interlace. See MIL-S-49301.

4.6.3 Scan direction. With the MX-10717(V)1/U operating in either the 2:1 side or the 2:1 rear interlace mode, the outputs of the azimuth and interlace position transducers (Scan-Interlace SM-D-773946) shall be observed with an oscilloscope. The output waveforms shall show the applicable shape and phase relationships shown in figure 1. Failure to meet the requirements of 3.6.3 shall constitute failure of this test.

4.6.4 Power consumption. See MIL-S-49301.

4.6.5 Scan efficiency. See MIL-S-49301.

4.6.6 Scan jitter in azimuth. See MIL-S-49301.

4.7 Environmental Tests. See MIL-S-49301.

6. NOTES

6.1 Intended use. The MX-10717(V)1/U is intended for use with the AN/ASQ-170, Target Acquisition and Designation Sight, and AN/AAQ-11, Pilot Night Vision Sensor (TADS/PNVS).

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:02Z
Check the source to verify that this is the currentversion before use.

6.2 Interchangeability. The MX-10717(V)1/U scanner built and tested in accordance with this specification and applicable drawings may be used in place of the MX-9872(V)1/UA scanner. The MX-9872(V)1/UA may not be used in place of the MX-10717(V)1/ scanner.

6.3 Subject Term (Keyword) Listing.

FLIR
Sensor
Scanner

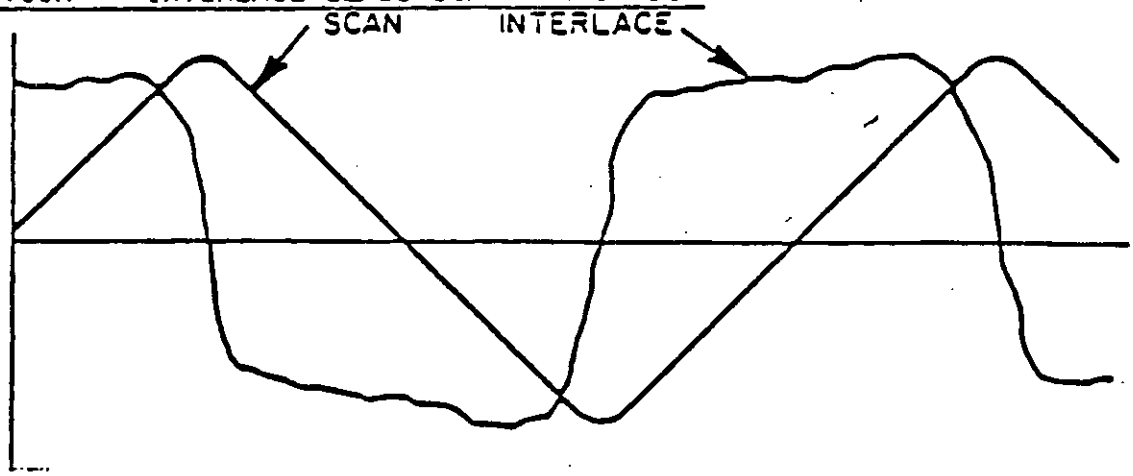
CUSTODIAN:
Army-CR

PREPARING ACTIVITY:
Army-CR
Project 5855-A290-1

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:02Z
Check the source to verify that this is the current version before use.

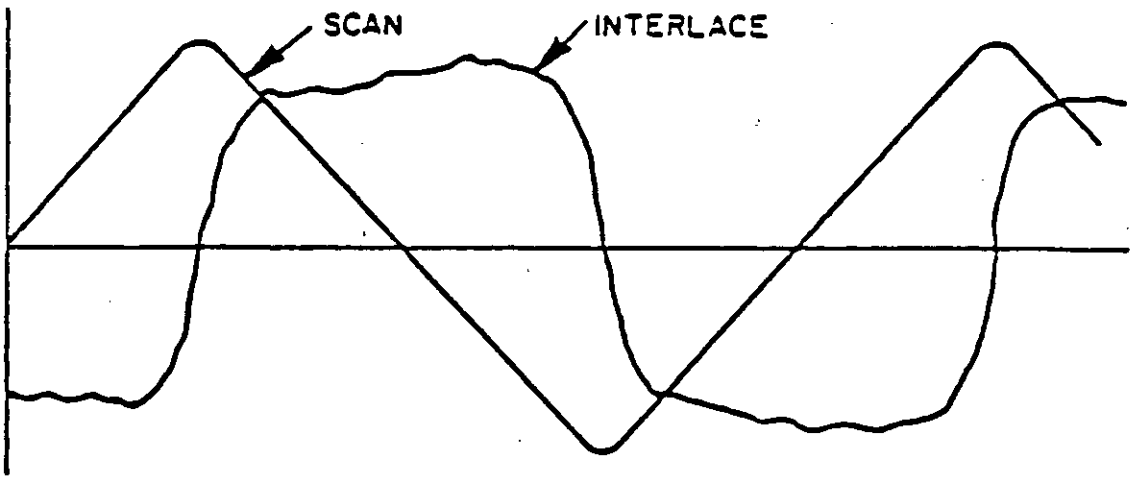
2:1 'REAR POSITION' - INTERLACE LEADS SCAN APPROX 90°

PL-1408/UA
P1: CONNECT
PIN 33 to 34
PIN 17 to 4
Oscilloscope:
(See Note)



2:1 'SIDE POSITION' - INTERLACE FOLLOWS SCAN APPROX 90°

PL-1408/UA
P1: CONNECT
PIN 31 to 33
PIN 17 to 4
Oscilloscope:
(See Note)



NOTE - CONNECT OSCILLOSCOPE AS FOLLOWS:

CHANNEL 1 - HIGH INPUT TO J3 (OR J2-7), LOW SCOPE TO GND.
CHANNEL 2 - HIGH INPUT TO J6 (OR J1-9), LOW SCOPE TO GND.

FIGURE 1. INTERLACE TRANSDUCER AND AZIMUTH POSITION TRANSDUCER OUTPUTS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:02Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-49301/1(CR)	Scanner, Mechanical, MX-10717(V)1/U

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE. HISA-FM 149-85

Source: https://assist.dla.mil – Downloaded: 2026-10-22T20:02Z
Check the source to verify that this is the current version before use.